

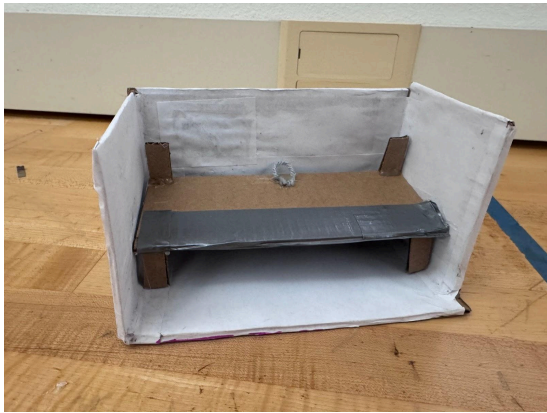
Glide Lounge Final Prototype Memo

Problem Statement

Michelle Harrison has a sofa bed at the end of her camper van that is supposed to convert from a sofa to a bed. The client has a difficult time moving between the two setups (sofa and bed) as the system is outdated—the planks are slipping apart from one another and switching positions requires disassembling the entire construction, which makes the practical use of it challenging. The client needs a solution that allows her to easily move the position from bed to couch and vice versa. The final prototype should also address shaking and noisiness, which arises from unevenness on the road while the trailer is moving.

Prior Prototypes

In the previous low-resolution prototype, the final design alternative was explored: two wood pieces with



hinges in between them, with a cord at the back to pull both planks upwards to access the storage space beneath. The prototype was low-resolution, as it used cardboard and hot glue. This method of construction allowed a faster decision to be made about the design because it was quick to construct, as well as benefited the overall budget of the project because no funding had to be used to construct the prototype. This scaled down prototype of the hinge design allowed major problems to be identified. Since the client wanted to access the generator in the storage space, both planks needed to fold up. However, in the fixed-leg design, the legs on the outer plank prevented folding. This

demonstrated that the legs for future designs must not impede folding. Another issue that came from this prototype is that even without legs, full range movement of the wall hinges was impeded by the outer plank. To solve this issue, it was decided to make the outer plank shorter than the inner plank.



The following low-resolution cardboard prototype of the leg mechanism was constructed after consulting with the client on how to resolve the issue of the impeding legs. The client said that they wanted the legs to be removable, which has been honored in the prototype. The prototype uses a screw mechanism that allows the legs to be screwed into and out of the plank, with the screw being embedded in the leg to ensure the screw is not lost. This prototype proved the screwing mechanism to be effective for the desired purpose, and was then incorporated in the high-resolution prototype.

Question

Does the prototype allow for quick and easy (requires at most two people and standard strength, process takes less than 3 minutes) conversion between bed and sofa positions?

The client's main issue with the previous sofa-bed design was that it was very difficult to shift between the sofa mode and the bed mode and vice versa. Therefore, minimizing the effort to shift positions was the team's primary objective. The main question regarding the ease of changing positions of the prototype, (with ease being defined as taking less than 3 minutes, requiring up to two people with average strength to set it up) will assess how well the primary objective is fulfilled by the prototype. Additionally, the prototype will allow the team to determine the alignment with the functions of allowing easy conversion between the bed and sofa systems and functioning effectively as both a bed and a couch. The metric scale for the force objective has been set to 0-5, where a score of 0 means that more than 50 pounds of force are required to use the design, while a 5 requires less than 10 pounds. For the time objective, the metric scale has been set to 0-4. A score of 0 means that it takes more than 150 seconds to convert between modes, while a score of 4 means that it takes less than 60 seconds. Answering this question will help determine how well the prototype fulfills the primary objective and whether any adjustments should be made to further increase the ease of adjustment between positions.

Prototype

Note: The prototype is not in the classroom because the evaluation plan included installing the prototype into the camper, where it now permanently resides, so photos are included throughout the memo to show all details of the design.

The prototype was first designed through multiple client consultations, low-resolution cardboard prototypes to brainstorm the possible mechanisms, and taking various measurements to determine the necessary size of the design in order to fit in the camper. After the prototype was ideated and tested on a smaller scale, a full-scale prototype was outlined and materials sourced. The convertible bed/couch system prototype was made by wood cutting, sanding, and drilling, and installing hinge mechanisms. Two planks were first acquired from Lowe's, and were then hand sanded to remove any splinters and create a polished finish. Long hinges and screws to fit were acquired from McMaster-Carr as well as smaller pieces of wood for the supportive legs that were obtained from the wood shop.

To determine the placement of supportive legs in the system, Finite Element Analysis was conducted. Since the front plank is connected to the back via a hinge and external ledge support, a small back support to replicate the ledge and two legs were placed to see how the pressure would be distributed. Each of the results include legs and the electrical box, which is hollow on the inside and part of the camper setup.

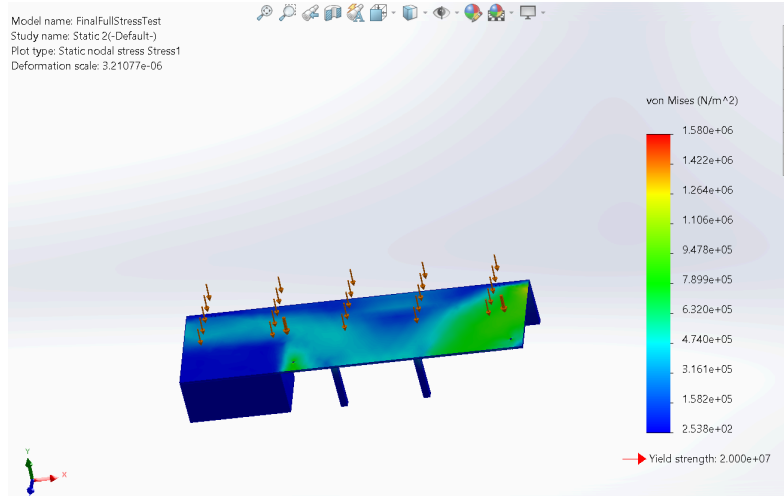


Figure 1: Back Support

The back ledge support was removed and the study was conducted with two legs and then three legs, roughly equally spaced. The updated pressure distribution can be seen in Fig. 2 and Fig. 3.

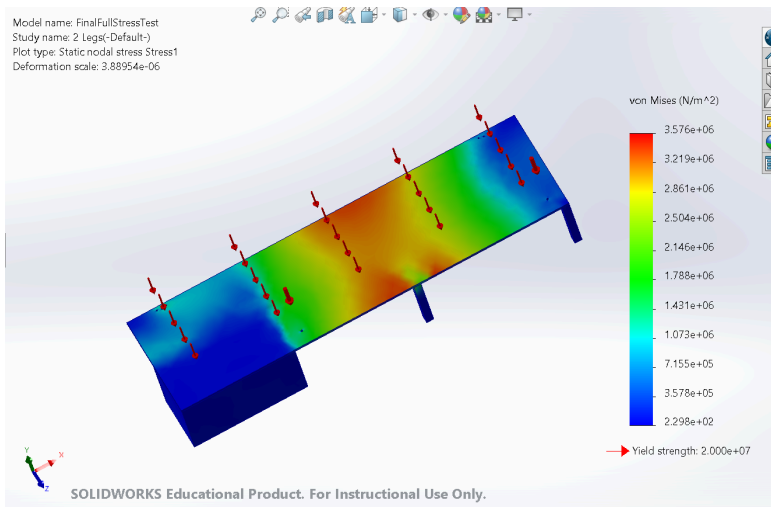


Figure 2. Two Legs, no back ledge support

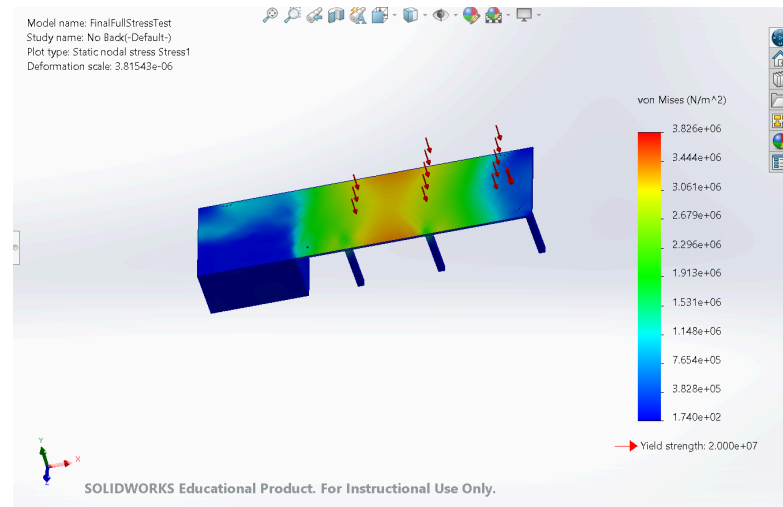


Figure 3. Three Legs, no back ledge support

As can be seen in the Von Mises graph, with two legs, there was an extreme amount of pressure in the middle. This pressure could cause the planks to break when a mass is placed on top, creating both a safety hazard and rendering the two-leg design difunctional. However, once a third leg is added in the middle, the pressure becomes more distributed, as can be seen in Fig 3. where there are no red high-pressure areas. Therefore, three legs were determined to be the best option for the prototype.

Once the number of supportive legs required was determined, they were cut out using a wood saw according to measured values from previously-acquired rectangular stocks of oak wood. The legs have

been designed so that they have a hole in the middle in which a screw is inserted in resin. The plank was tapped with wood inserts that fit those screws so that the legs can be screwed on and off as desired by the client. This allows ease of use whenever the position of the system is changed, as well as comfort, since the legs can be removed when the mechanism is in couch mode (they would be pointing upwards otherwise when the system is hinged). Holes were drilled in the bottoms of the legs, and grub screws were inserted (Figure 6). Afterwards, three hinges were installed to connect the two planks, and additional hinges were added to the opposite edge of the plank without legs. These hinges were connected to the back wall of the camper via screws (Figure 4). The prototype features a high-resolution manufacture that allows it to most closely reflect the desired design and illustrate the ideas previously expressed through low-resolution cardboard prototypes. Since it is close to the final design, it can be evaluated based on its effectiveness in fulfilling the primary objective as outlined by the main testing question. The prototype fulfills all of the project constraints. The final design cost less than \$150, and was installed successfully into the van, showing that it fits into the camper. During testing the prototype only needed one person to convert it from bed to couch mode, and the conversion took on average 82 seconds. Additionally, the user was able to lift the design to access the storage space beneath, so all constraints were met or exceeded.



Figure 4. Installed Couch Form



Figure 5. Installed Bed Form



Figure 6. Legs with Inserted Screws



Figure 7. Insertion of Screws



Figure 8. Wood Insert Locations



Figure 9. Installation of Hinges Between Planks

Evaluation Plan

1st Evaluation—time to set up

- 1) Switch prototype from bed mode to couch mode, and then back to bed mode. This should include starting with the design fully set up as a bed, then removing the cushions, placing them aside, folding up the outside plank, removing the legs, and putting back the cushions in their couch orientation. Going back to bed mode should be the same steps, just in the reverse order.
- 2) Record the amount of time it takes to perform this cycle. Timed using a stopwatch and two people, one to time and one to convert the prototype.
- 3) Repeat steps 1 and 2 five times and calculate the average time. Things that should be recorded during this testing include the amount of time taken, as well as if any difficulties occur during a particular step of the conversion cycle. These difficulties should be noted so that in a future design iteration, changes can be made to eliminate them and decrease the conversion time. The user converting the design should also record their feelings on which step in the cycle takes the longest, so that increasing the efficiency of that step could be taken into consideration for a future iteration as well. Future and prior iterations with the same main objective - simplifying the process to switch between couch and bed - can use the same evaluation plan to test the prototype.

2nd Evaluation—Weight of conversion

- 1) Record the amount of weight that needs to be lifted (in pounds) in order to switch between bed and couch (with qualitative feedback). For this step, the user should first pick up a 20, 30 and 50 pound dumbbell, to get a sense for what each feels like. The user should then convert the system between bed and couch and record which amount of weight feels most similar. The qualitative feedback that should be recorded at this step includes asking the user if they feel that they are over-exerting themselves while using the design, or if it feels easy.
- 2) Record the amount of weight that needs to be lifted (in pounds) in order to switch between couch and bed (with qualitative feedback) The user should complete the same process of picking up the weights and comparing the feel of the weights to the amount of weight required to lift to shift

between the two modes. The qualitative feedback that should be recorded at this step includes asking the user if they feel that they are over-exerting themselves while using the design, or if it feels easy.

- a) Note: Steps one and two were estimated using this subjective method of measuring due to the fact that no portable scale devices were available at the time of testing.
- 3) Repeat steps 1 and 2 five times and calculate the average estimated weight. For each trial, the team lifted the prototype at different locations along the edge of the plank to better estimate the average force. Future and prior iterations with the same main objective - simplifying the process to switch between couch and bed - can use the same evaluation plan to test the prototype.

The evaluations will be scored based on the following metrics:

Minimize set up time	Time needed to convert from bed to couch and back	Score
	More than 150 seconds	0
	121 seconds - 150 seconds	1
	91 seconds - 120 seconds	2
	61 seconds - 90 seconds	3
	Less than 60 seconds	4

Figure 1: Metric for time

Minimize Force to Shift Positions	Weight needed to lift to shift positions (in Pounds)	Score
	51+	0
	41-50	1
	31-40	2
	21-30	3
	11-20	4
	<10	5

Figure 2: Metric for Force

Evaluation Results

From the prototype testing, several important results were gathered, including the amount of weight to lift the design to convert between bed and couch mode, as well as how quickly the user is able to transition between modes. The first result gathered from the testing was that the prototype is able to be quickly converted from one mode to the other by one user, with quickly being defined as less than 150 seconds. The average amount of time it took to shift the prototype from bed mode to couch mode and back was 82 seconds. This included removing the decorative cushions, attaching/removing the legs, and using the hinges to flip the planks from one mode to the other. When graded against the metric scale, the prototype scored a 3 out of 4 on the time metric, and is well below the cutoff for the setup time constraint of 7 minutes. The other data that was recorded included the user stating that they felt the step that took the longest during setup was screwing and unscrewing the legs from the plank. In any future iterations of this prototype, the only improvement that could be made to decrease the amount of time required to set up would be to have a different mechanism to attach the legs to the plank instead of screws, since the action of screwing/unscrewing all three legs to the threaded inserts takes up the majority of the set up time.

The second result gathered from the testing was that the prototype can easily be converted from one mode to another by a singular user, since it does not require an extreme amount of force. During testing, the maximum weight required to be lifted/exerted by the person setting up at a singular moment when converting from bed to couch and couch to bed was estimated to be 20 pounds. During evaluation, the user said that they felt they did not have to over-exert themselves in order to convert between modes, and that it felt easy. When grading the results against the metric scale, the prototype scored a 4 out of 5 on the force metric and was successful in achieving the force constraint, indicating that the design is easy to convert between bed and couch modes. In future evaluation testing of this prototype, it would be ideal to have a scale or device that would be able to quantitatively measure the amount of weight required to be lifted to shift modes, since that would give an objective measurement that could be graded by the same metric scale. For future iterations of this prototype, the design could be improved by using a different material for the planks that is more lightweight, since most of the weight that required the user to use force to lift was due to the wooden planks. If a different material was used, much of the weight of the design would be reduced and the user would need to use even less force, which would be ideal. Moving forward from this prototype, ideally the client will be able to use this prototype for the foreseeable future, since it is a successful solution to the client's problem, fulfilling all constraints and converting between bed and sofa mode quickly and easily, so theoretically no further action is required moving forward. However, if another iteration of the design were to be created, the improvements that could be made include changing the material of the planks to be more lightweight, as well as changing the method of attachment for the legs in order to be faster to attach. Additionally, the first low-resolution prototype included a cord attached to the planks to create an easier way to lift the planks up to access the storage space, and the final prototype did not include this. This was not included because the client had a specific strap that they wanted to use, so it was decided that they would add the strap that they wanted after the prototype was installed. Future iterations of the prototype could be improved by adding this cord to the design so that it could be used during the evaluation stage to measure the amount of force required to lift using the strap.